Uni. Roll No:.....

**SET – C**

**GLA University, Mathura**

**Course: -B.Tech. (CSE) II – Year, I – Mid Term (Even Sem.) Examination, 2012 – 13**

**Subject: - Database Management System (ITE 221)**

**Time:-90 Minutes** **M.M:-20**

Notes:-

1. Answer all questions from Group A (which is) compulsory, Any Two from Group B and Any Two from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

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**Group – A (All Questions are Compulsory)** **08 X 0.5 = 04**

**I** Data is a:
- Metadata,
- Information only,
- Raw fact & figure,
- Query

**II** An E–R diagram that shows the concepts of Specialization & Generalization is known as:
- Effective E – R diagrams
- Expanded E – R diagrams
- Extended E – R diagrams
- Enhanced E – R diagrams

**III** What is the cardinality of a table with 50 rows and 5 columns:
- 5
- 250
- 50
- 10

**IV** What is the degree of a table with 40 rows and 4 columns:
- 4
- 160
- 40
- 10

**V** The column in the child table that references a primary key of the parent table is called as:
- Foreign key
- Control Key
- Candidate Key
- Composite key

**VI** A relationship between a weak entity set is called as the:
- Identifying relationship
- Connecting relationship
- Weak relationship
- Associative relationship

**VII** To sort records we use:
- HAVING clause
- GROUP clause
- ORDER BY clause
- WHERE clause

**VIII** INSERT, DELETE & UPDATE commands are put under:
- DML
- DCL
- DDL
- DQL

**Group – B (Answer any TWO)** **03 X 2 = 06**

**I** Differentiate between Physical and Logical Data Independence.

**II** Explain the roles & responsibilities of a DBA.

**III** Define Primary Key, Foreign Key and Super Key.

**Group – C (Answer any TWO)** **05 X 2 = 10**

**I** Explain the Three – Level architecture of DBMS in detail with the help of diagram.

**II** Given the following relational schema:
sailors (sid, sname, rating, age), boats (bid, bname, color), reserves (sid, bid, day)
Find the relational algebra expression for the following:

1. Find names of sailors who have reserved boat #103.
2. Find names of sailors who have reserved a red boat.
3. Find sailors who have reserved a red or a green boat.
4. Find sailors who have reserved a red and a green boat.
5. Find the names of sailors who have reserved all boats.

**III** Construct an ER diagram for a Hospital Management System.

Uni. Roll No:-.....

# SET-D

## GLA University, Mathura

**Course: -B.Tech. II-Year, I-Mid Term (Even Sem.) Examination, 2012-13**

**Subject: - Operating System (CSE-221)**

**Time:-90 Minutes** **M.M:-20**

**Notes:-**

1. Answer all questions from Group A (which is) **compulsory**, **Any Two** from Group B and **Any Two** from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

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**Group-A (All questions are compulsory)** **08X 0.5=04**

**1. Which of the following does not form an OS LAYER?**

- **Application Programs**
- **Critical Section**
- **Shell**
- **Kernel**

**2. A small program which loads OS into the memory is called as**

- **BIOS**
- **ROM**
- **None of the above**
- **Bootstrap Program**

**3. IPC can be done through-**

- **Messages**
- **Mails**
- **System calls**
- **Traps**

**4. Moving a process from main memory to the disk is called as**

- **Caching**
- **Swapping**
- **Spooling**
- **Scheduling**

**5. Information about a process is maintained in a _____.**

- **Translation Look Aside Buffer**
- **Program Control Block**
- **Process Control Block**
- **Stack**

**6. In the blocked state**

- **the process which is running is found**
- **The processes waiting for I/O are found**
- **the process waiting for processor is found**
- **None of the above**

7. Switching the CPU to another Process requires saving state of the old process and loading new process state is called as _____.

- Process Blocking
- Context Switching
- Time-Sharing
- None of the above

8. A thread

- Is a lightweight process where context switching is high.
- Is used to speed up paging.
- Is a lightweight process where context switching is low.s
- None of the above

## Group-B (Answer any Two) 3x2=6

Q1.What are the main advantages of the Micro Kernel approach to system design? Draw its structure also.
Q2. Explain different types of schedulers.
Q3. Explain process and its life cycle with the help of diagram.

## Group-C (Answer any Two) 5x2=10

Q1.Explain Operating System .List and explain the services provided by the operating system.
Q2. How IPC will take place using Shared Buffer and Message Passing.
Q3. How would you implement threads in different ways?

# GLA University, Mathura

**SET D**

## I-Mid Term Examination, 2012-13

**Course: - B.Tech.** **II-Year, IV-Sem.**

**Subject: - Computer Graphics (CSE 223)** **Uni. Roll No:-**

**Time:-90 Minutes** **Total Marks:-20**

**Notes:-**

1) Answer all questions, Section A is **compulsory**, **Any Two** from Section B and **Any Two** from Section C.
2) All parts of a question (I, II, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

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## SECTION-A

**Objective Questions: (0.5*8 = 04)**

I. Pixel phasing is a technique for:

- LCD
- anti-aliasing
- hidden line removal
- Shading

II. LED is which type of display?

- Impact
- Non Emissive
- Refresh
- Emissive

III. A computer with 32-bit word has:

- 32 bytes /word
- 2 bytes/word
- 8 bytes/word
- 4 bytes/word

IV. Frame buffer is used to store

- Number of pixels in the image
- co-ordinate values of image
- image definition
- intensities of pixels

V. Beam penetration method can produce maximum colors

- 2
- 5
- 4
- 3

VI. DDA Stands for

- Digital Differential Algorithm
- Digital Analyzer Differential
- Digital Data Algorithm
- Digital Differential Analyzer

VII. Boundary Fill algorithm is the simplest_____filling algorithm

- Line
- Volume
- Frame
- Area

VIII. Slop of the line joining the points (1, 2) and (3, 4) is

- 3
- 2
- 0
- 1

## SECTION-B (Attempt any two)

I. A laser printer is capable of printing two pages (size 9x11 inch) per second at resolution of 600 pixels per inch. How many bits per second does such device require? (3)

II. Explain the working of CRT. Distinguish between Raster and Random scan devices (1.5+1.5)

III. Distinguish between Flood Fill and Boundary Fill Algorithms and explain any one of them. (1+2)

## SECTION-C (Attempt any two)

I. Write Cohen Sutherland algorithm for Line Clipping. Use Cohen Sutherland line clipping algorithm to clip the line **P1 (70, 20) to P2 (100, 10)** against the window having diagonally opposite corners as **(50, 10)** and **(80, 40).** (5)

II. Write Midpoint Circle Drawing algorithm and execute the same for radius **r =10** and centered at **(0, 0).** (5)

III. Write Bresenham's line drawing algorithm. Consider a line from (20, 10) to (30, 18) by using Bresenham's line drawing algorithm rasterizing this line. (5)

**GLA University, Mathura** **SET-B**

**I-Mid Term Examination, 2012-13**

**Course: - B.Tech (CSE)** **II-Year, IV-Sem.**

**Subject: - Introduction to Microprocessor (CSE-222)** **Uni. Roll No:-**

**Time:-90 Minutes** **Total Marks:-20**

**Notes:-**

1) *Answer all questions; Section A is compulsory, Any two from Section B and Any two from Section C.*
2) *All parts of a question (a, b, etc.) should be answered at one place.*
3) *Answer should be brief and to-the-point and be supplemented with neat sketches.*
4) *Any missing or wrong data may be assumed suitably giving proper justification.*
5) *Figures on the right-hand side margin indicate full marks.*

## SECTION-A

**Objective Questions** (Write the correct answer): **(8*0.5 = 4)**

I. What is the name of machine cycle if IO/$\overline{M}$ =1, $S_1$=0 and $S_0$=1?
- Memory Read
- IO Read
- OPCODE-Fetch
- IO Write

II. Which of the following equations is correct?
- Microprocessor=Microcomputer + Memory Unit + IO devices
- IO devices= Microprocessor-Memory
- Microcomputer=Microprocessor + Memory Unit + IO devices
- None of these

III. For proper functioning microprocessor needs .....
- Clock
- Control Unit
- Count
- Micro-program sequencer

IV. Ready pin of a microprocessor is used
- To indicate that the microprocessor is ready to receive inputs.
- To indicate that the microprocessor is ready to receive outputs.
- To provide direct memory access
- To introduce delay during execution.

V. Which of the following statement is true?
- A machine cycle consists of one or more instruction cycle.
- An instruction cycle is made up of machine cycles and a machine cycle is made up of number of T-states.
- The group of machine cycle is called a T-state.
- None of the above

VI. Which Microprocessor pins are useful for request and acknowledge DMA transfer?
- HOLD and HLDA
- Ready and wait
- Reset and ready
- None of these

VII. First modern computer was called
- Flow-Matic
- UNIVAC-I
- ENIAC
- Intel

VIII. Difference engine could perform .....operation.
- Addition and Subtraction
- Addition
- Counting
- Subtraction

**SECTION-B (Attempt any two)** **(3*2=6)**

I. Write 8085 instruction for
   (a) Load 2010H data to HL Register Pair.
   (b) Transfer contents of 1050H memory location to B Register.
   (C) Load 40H to A register.
   (d) Copy contents of H register to L register.
   (e) Add contents of accumulator with accumulator.
   (f) Add 40H as immediate data with contents of accumulator.

II. Explain the need to DEMULTIPLEX the bus $AD_7$-$AD_0$ with neat diagram.

III. How microprocessor [redacted] fetch OPCODE from memory? Draw the timing diagram of OPCODE Fetch from memory to microprocessor unit.

**SECTION-C (Attempt any two)** **(5*2=10)**

I. Write an assembly language program for exchange the contents of memory locations 1000H and 2000H with proper description of each instruction.

II. Draw architecture of basic microprocessor and explain its all components. Write names of addressing modes used in 8085 microprocessor.

III. Draw the pin diagram of 8085 and specify the function of each pin separately.

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**I-Term Examination – B.Tech-II Year** **SET-A**

**Session 2012-2013 (Even Semester)**

**Subject: Computer Networks**

**Time: 90 Mins.** **Max. Marks: 20**

**Notes:-**

1. Answer all questions from Group A (which is) **compulsory**, **Any Two** from Group B and **Any Two** from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

## *Section – A*

***Attempt all the questions* (08X 0.5=04)**

**1. ISDN stands for**

- **Integral Service Dynamic Network**
- **International Subscriber Dialup Network**
- **International Service Digital Network**
- **Integrated Service Digital Network**

**2.In OSI network architecture, the dialogue control and token management are responsibility of**

- **Session layer**
- **Network layer**
- **Transport layer**
- **Data link layer**

**3.An example of an analog communication method is**

- **Laser beam**
- **Microwave**
- **Voice grade telephone line**
- **All of the these**

**4.Which of the following performs modulation and demodulation?**

- **Fibre optics**
- **Satellite**
- **Coaxial cable**
- **Modem**

**5.The process of converting analogue signals into digital signals so they can be processed by a receiving computer is referred to as:**

- **Modulation**
- **Demodulation**
- **Synchronizing**
- **Digitising**

**6.Which of the following is an advantage to using fiber optics data transmission?**

- **Resistance to data theft**
- **Fast data transmission rate**
- **Low noise level**
- **All of above**

**7.Which of the following communication modes support two-way traffic but in only one direction at a time?**

- **Simplex**
- **Half duplex**
- **Three-quarters duplex**
- **All of the above**

**8.The loss in signal power as light travels down the fibre is called**

- **Attenuation**
- **Propagation**
- **Scattering**
- **Interruption**

## Section – B *[Attempt any Two questions. (03x02=06)]*

**Q1: Explain the trade-offs between circuit-switched and packet-switched networks. Give suitable example for each type of network.**

**Q2: Draw the graph of the NRZ-L, Manchester, Differential Manchester Encoding Scheme for 00110011 data stream. Assuming that the last signal level has been positive.**

**Q3: What signal-to-noise ratio (in dB) is needed to put a TI carrier on a 50-KHZ line? Also compute the number of signal level.**

## *Section – C [Attempt any Two questions ( 05x02=10)]*

**Q1: What is ISDN? Explain in detail about various types of functional groups and reference points used in ISDN?**

**Q2: A path in a digital circuit-switched network has a data rate of 1 Mbps. The exchange of 1000 bits is required for the setup and teardown phases. The distance between two parties is 5000 km and propagation speed is $2x10^8$ m/sec. What is the total delay if 1000 bits of data are exchanged during the data transfer phase?**

**Q3: Write the various functions and services of all OSI layers in detail.**

**-:All the Best:-**

SET-A

Uni. Roll No:-.....

# GLA University, Mathura

**Course: -B. Tech. II-Year, I-Mid Term (Even Sem.) Examination, 2012-13**

**Subject: - Environmental Studies (CEE-201)**

**Time:-90 Minutes** **M.M:-20**

**Notes:-**

1. Answer all questions from Group A (which is **compulsory**), **Any Two** from Group B and **Any Two** from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

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## Group-A (All questions are compulsory) 08X 0.5=04

**It contains 8 questions. Choose and write the correct answer.**

I. Who lead the Tehari Bachao Andolan -

- Sh Sundrlal Bahuguna
- Smt. Medha Ptekar
- Ms Arundhati Ray
- Rajendra Singh

II. Who is popularly known as green judge in India-

- Sh M.C. Mehta
- Sh Anil Aggrawal
- Sh Salim Ali
- Kuldeep Singh

III. Which one is the ecomark of India-

- ISO 14000
- Earthen pitcher
- Globe with a tree
- Cloth bag

IV. Which can be replenished in the nature-

- Bio fuel
- Petroleum
- Coal
- Nuclear fuel

V. World wet land day is celebrated on-

- February 2
- March 21
- March 22
- March 23

VI. Which ecosystem has the highest productivity-

- Lentic
- Lotic
- Estuarine
- Marine

VII. Global atmospheric temperatures are likely to increased due to-

- burning of fossils fuel
- water pollution
- soil erosion
- none of the above

VIII. The biomass pyramid of aquatic ecosystem is-

- Upright
- Inverted
- Both
- None

P.T.O.

## Group-B (Answer any Two) 3x2=6

I Why afforestation is needed?

II Differentiate conservation of resources and sustainable development.

III Write short note on multidisciplinary nature of environmental science.

## Group-C (Answer any Two) 5x2=10

I Discus the vertical structure and function of atmosphere and show related phenomenon.

II Define ecosystem. Give an account of structure and name function of an ecosystem

III What is fluoride problems in drinking water? Describe the sources, effects of excessive intake and solution measures of it.